

CSE220 Theory Lecture Plan				
Class	Topic	Detail Instructions	Quiz	Remarks
1	1. Importance of Data Structure Course. 2. Data Structure Syllabus Discussion. 3. Complexity Primer	1. Worst case time and space complexity (Big O notation only) 2. Growth of function. Draw chart for function. Example function: Binary search and Linear search graph for time complexity. 3. For space give example reading a file word by word (close to constant space) reading a file fully before operation (close to file length n) Materials - Chapter 0		
2	2. 2D array Basics and Operations	1. What is 2D array 2. Creation iteration of 2D array 3. Basic Operations. E.g: summation, diagonal sum, printing 4. Matrix multiplication 5. Index conversion between 1D and multidimensional array Materials - Chapter 1		
3	3. Linked List Primer	1. Link List Node Construction. 2. Link List Traversal. 3. Linked List with only Head and then with Both Head and Tail. 4. Append and Prepend in Linked List. Materials - Chapter 2		
4	4. Linked List Operations	1. NodeAt 2. Insert, remove at particular index. 3. Linked List Rotation. Materials - Chapter 2		
5	5. List Variations (Class 1)	1. Introduce Double Linked List 2. Show Insert-remove in Double Linked List Materials - Chapter 2		
6	6. List Variations (Class 2)	1. Circular Linked Lists 2. Dummy Headed Linked List Materials - Chapter 2		
7	7. Hashing and Hashtable	1. Explain hash functions. 2. Discuss Hash table implementation for forward chaining based implementation. 3. Show insert, delete operation for the discussed implementation.		

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8	8. Stack	1. Show stack implementation using linked list 2. Solve one or two programming problems using stacks among: (parenthesized expression validation, postfix arithmetic expression evaluation)		Only Linked List based Stack
9	9. Queue	1. Show Linked List Based Implementation. 2. Some problem solving using queue. E.g: remove odd number from a queue 3. Array based Queue		Both Linked List and array based implementation
10	10. Problem solving 11. If there is any lag	Cover any left out topic and solve problem		
11	12. Review Class			
12	13. Tree Terminologies	1. Discuss basic concepts. 2. Recursion recap (will be done in lab) 3. Discuss importance of recursion in tree programming		
13	14. Binary Tree 1	1. Binary Tree characteristics and creation using Array. 2. Tree Traversal 3. Types of Binary Tree		
14	15. Binary Tree 2 and BST 1	1. Binary Tree Coding (height, depth, level etc.) 2. Characteristics of BST		
15	16. BST 2	1. BST insertion and deletion 2. BST balancing		
16	17. Heap 1	1. Heap characteristics (Max and Min) 2. Insertion and deletion		
17	18. Graph 1	1. Graph terminology and types (weighted, unweighted; directed and undirected) 2. Graph representation (Adjacency Matrix and List)		
18	19. Graph 2	1. Graph programming of adjacency matrix and list) 2. Structuring a graph node for traversal (E.g: Adding the visited attribute) 3. Representation of weighted graph using adjacency list (use different edge class for connection)		
19	20. Graph traversal	1. Simulation and pseudocode / Code of BFS 2. Short exercise of BFS to DFS conversion		